Computer Science & IT



Data Structure

Tree

Lecture No. 01



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Recap of Previous Lecture



Topic

Topic

Topic

Topic

Topic

Circular Linked

queue using linked list

Slide

Topics to be Covered



Topic	Tree basic
Topic	Tree Terminology
Topic	Rooted tree
Topic	forest
Topic	k-ary tree

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Topic : Tree

- Tree
- Application
- Definition
- Rooted tree
- Terminology
- Forest
- Number of nodes in forest

Modeling Related

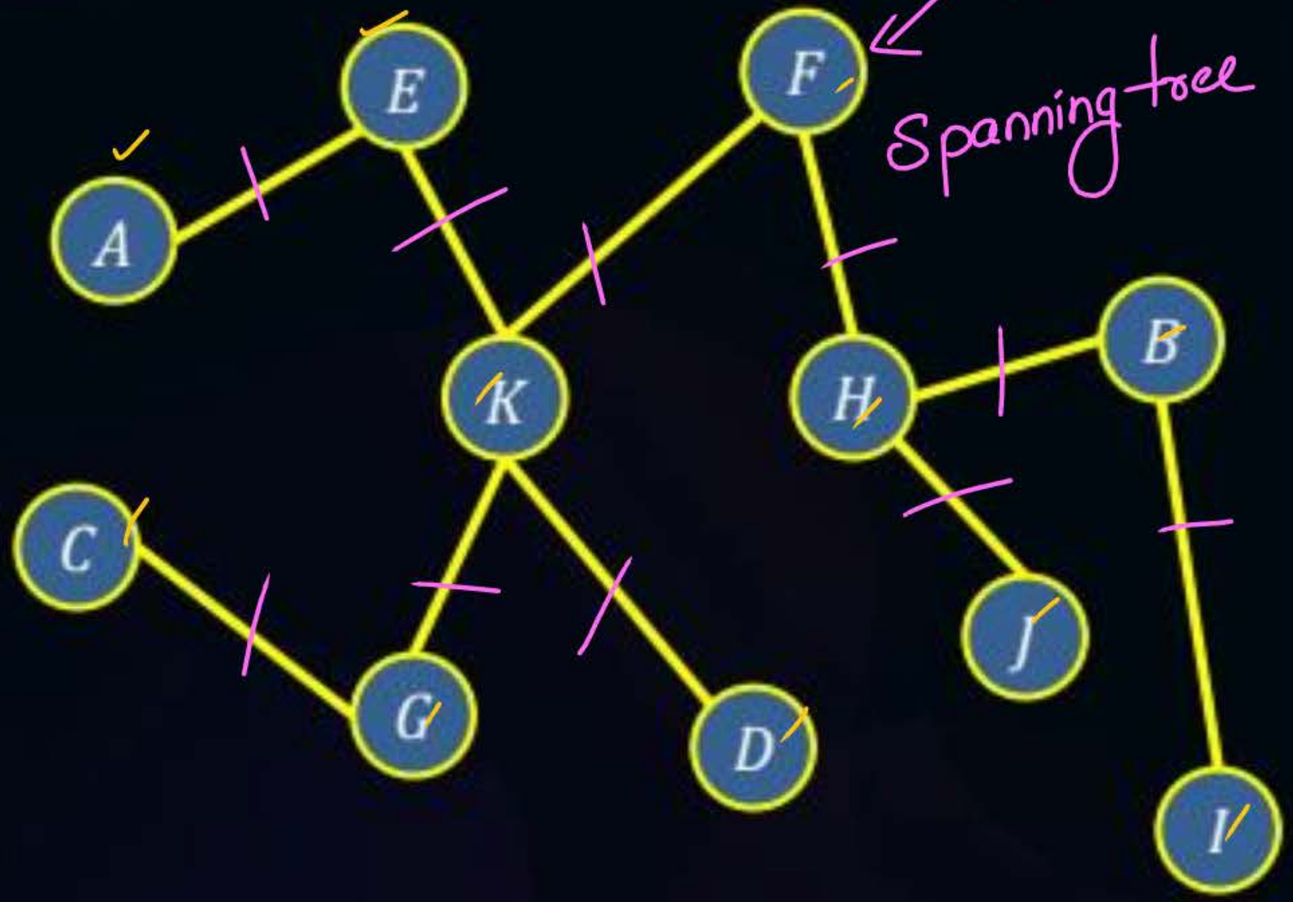
Graph : Graph is set of vertices & edges

$G(V, E)$

Edges between pair of vertex

Slide

Topic : Tree



Tree is a special graph

1 Tree is connected graph

2 Tree is an acyclic graph

graph without any cycle

3 A tree always consists of n vertices

& n-1 edges

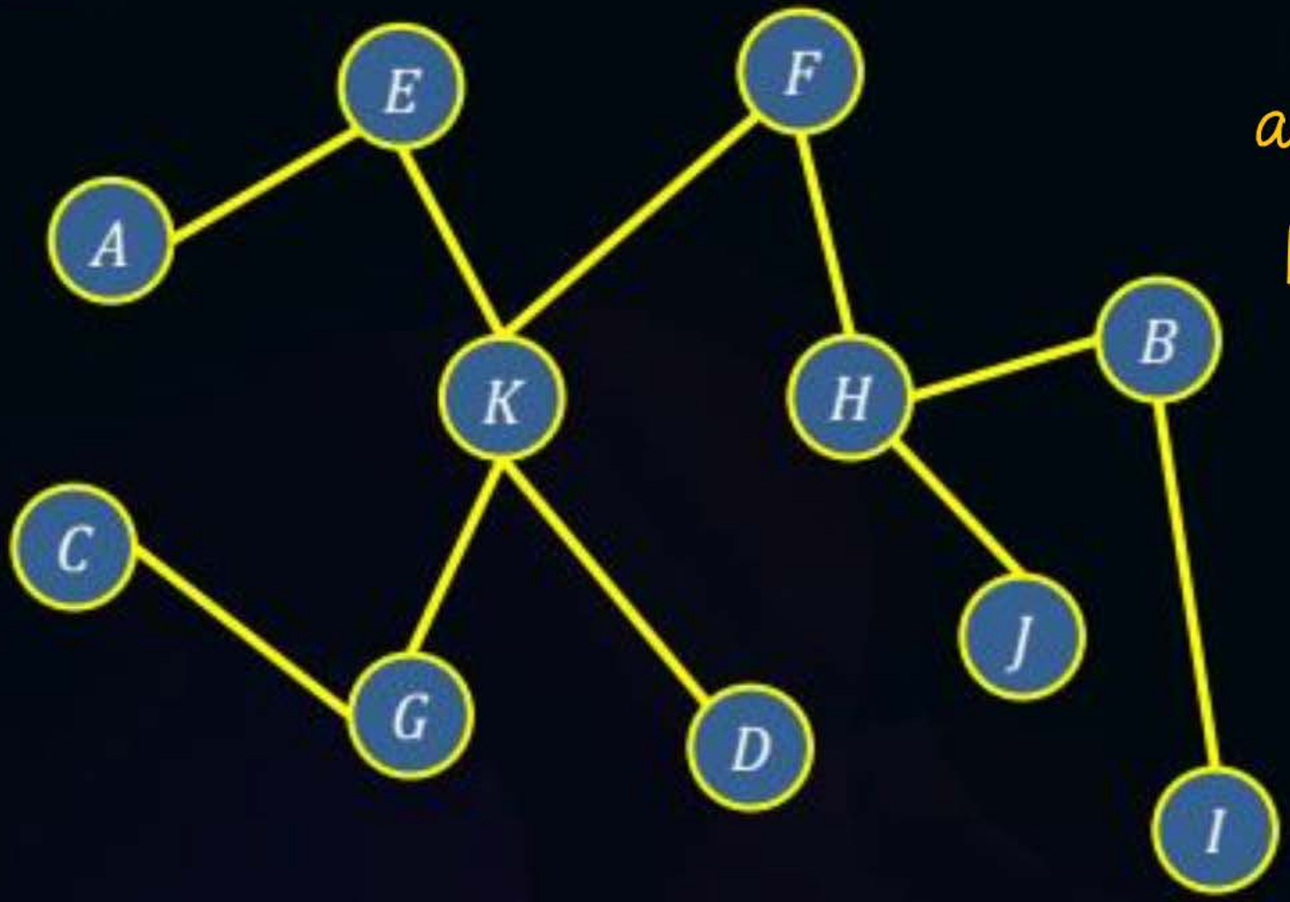
n = 11

edges n-1 = 11-1 = 10}

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Topic : Tree



Rooted tree :

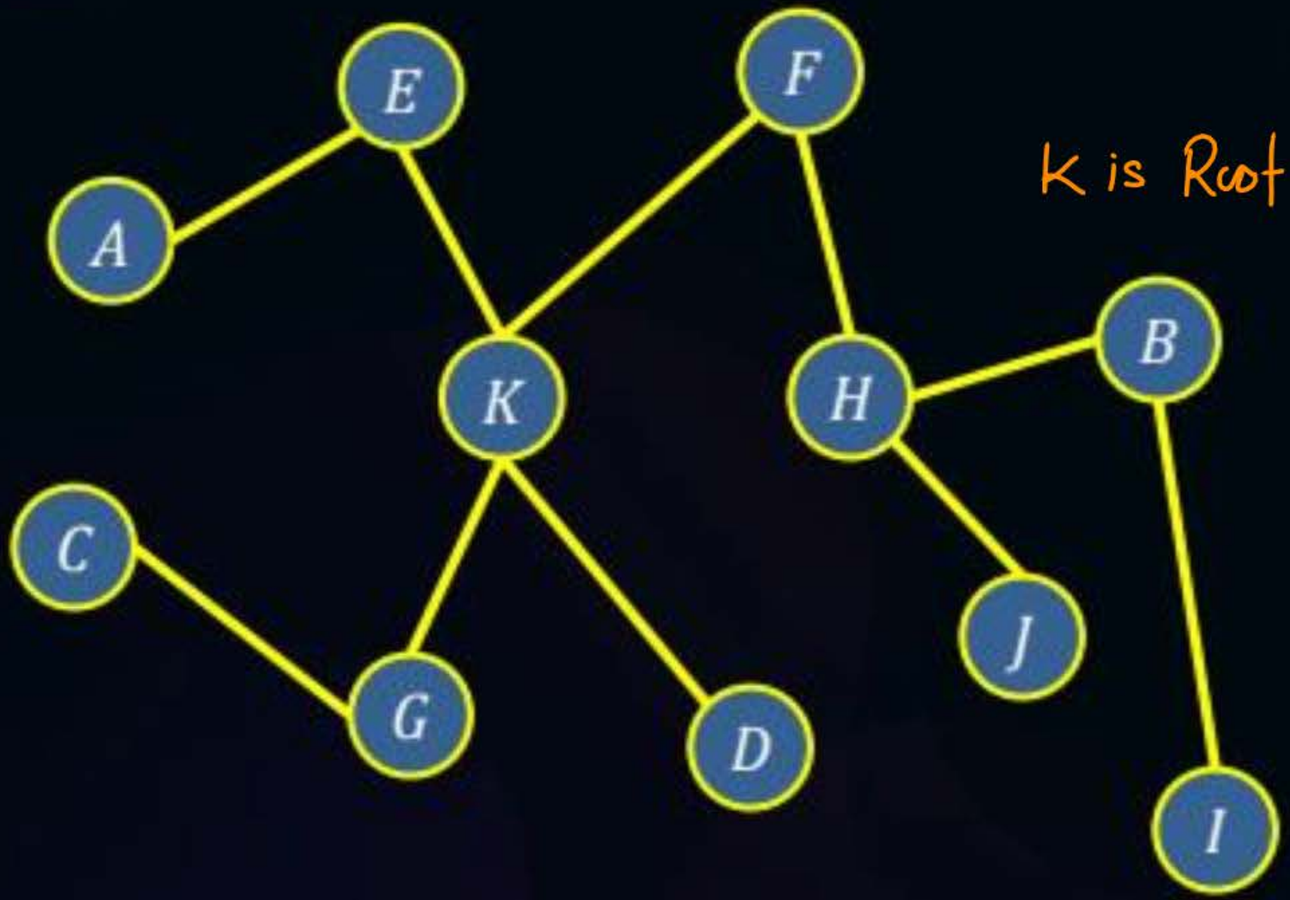
one of distinguished Node selected as Root and Rest node organized as parent & child.

Root having no parent.

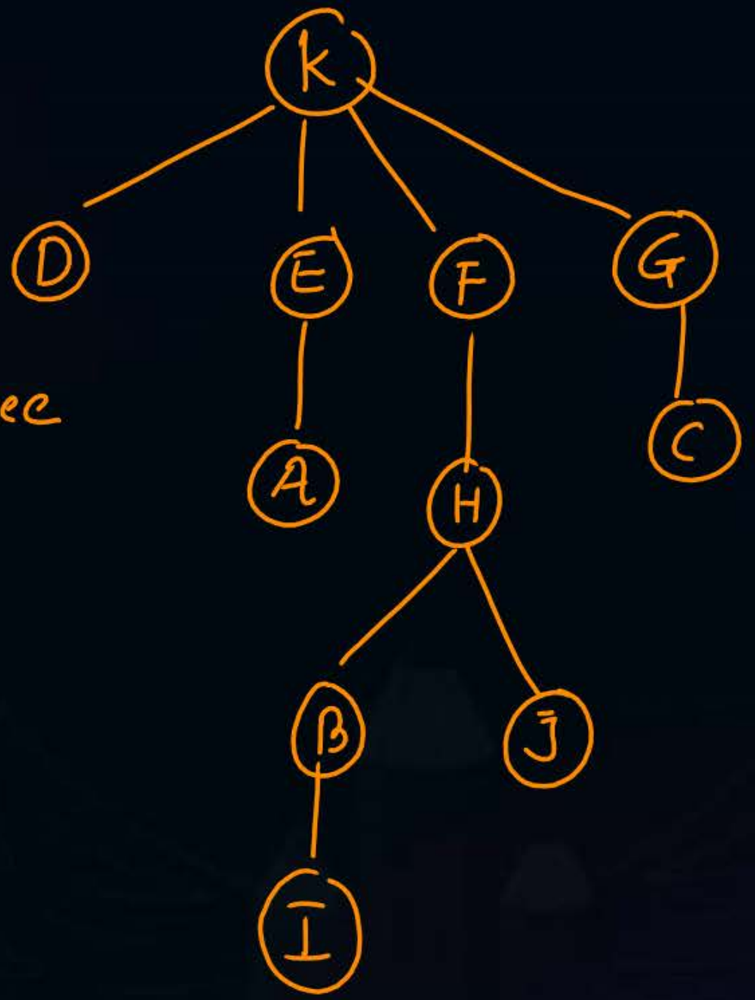
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Topic : Tree



K is Root of tree

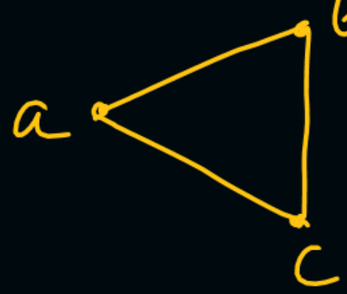


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Simple graph

No multiple edges

and No loops



a-b-c-a





disconnected



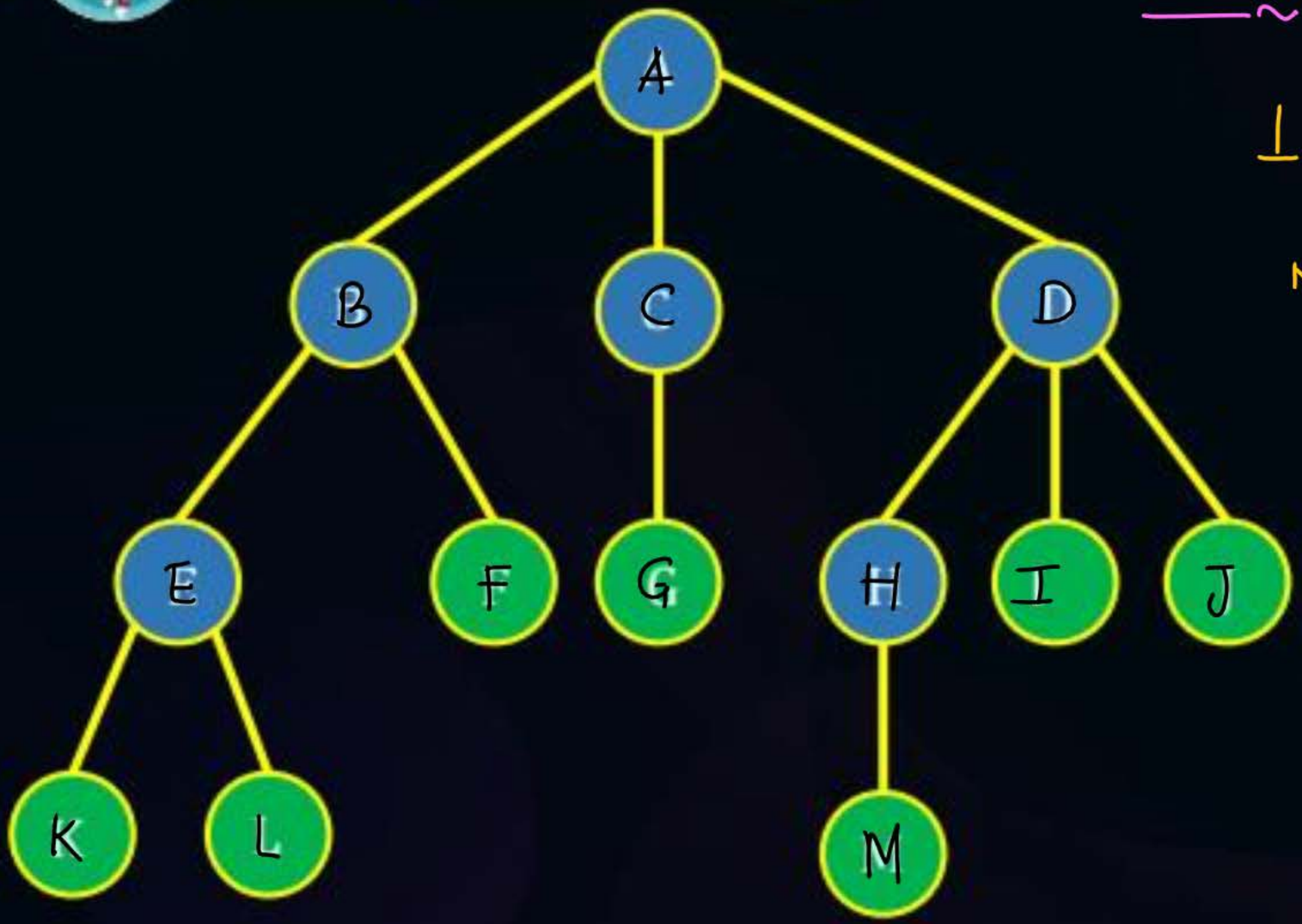


Topic : Tree

Terminology

Leaf Node : A Node having

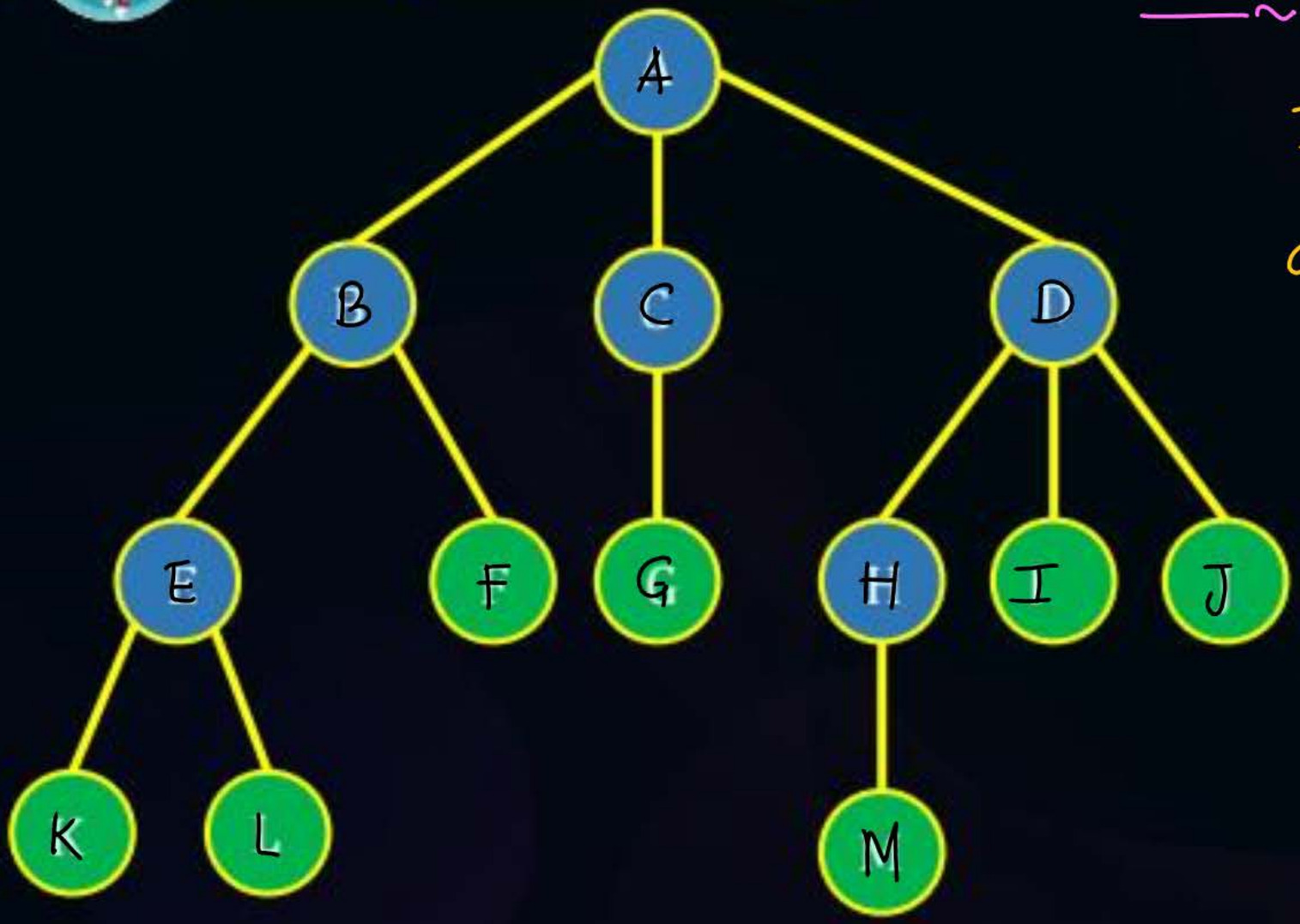
No children



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Topic : Tree



Terminology

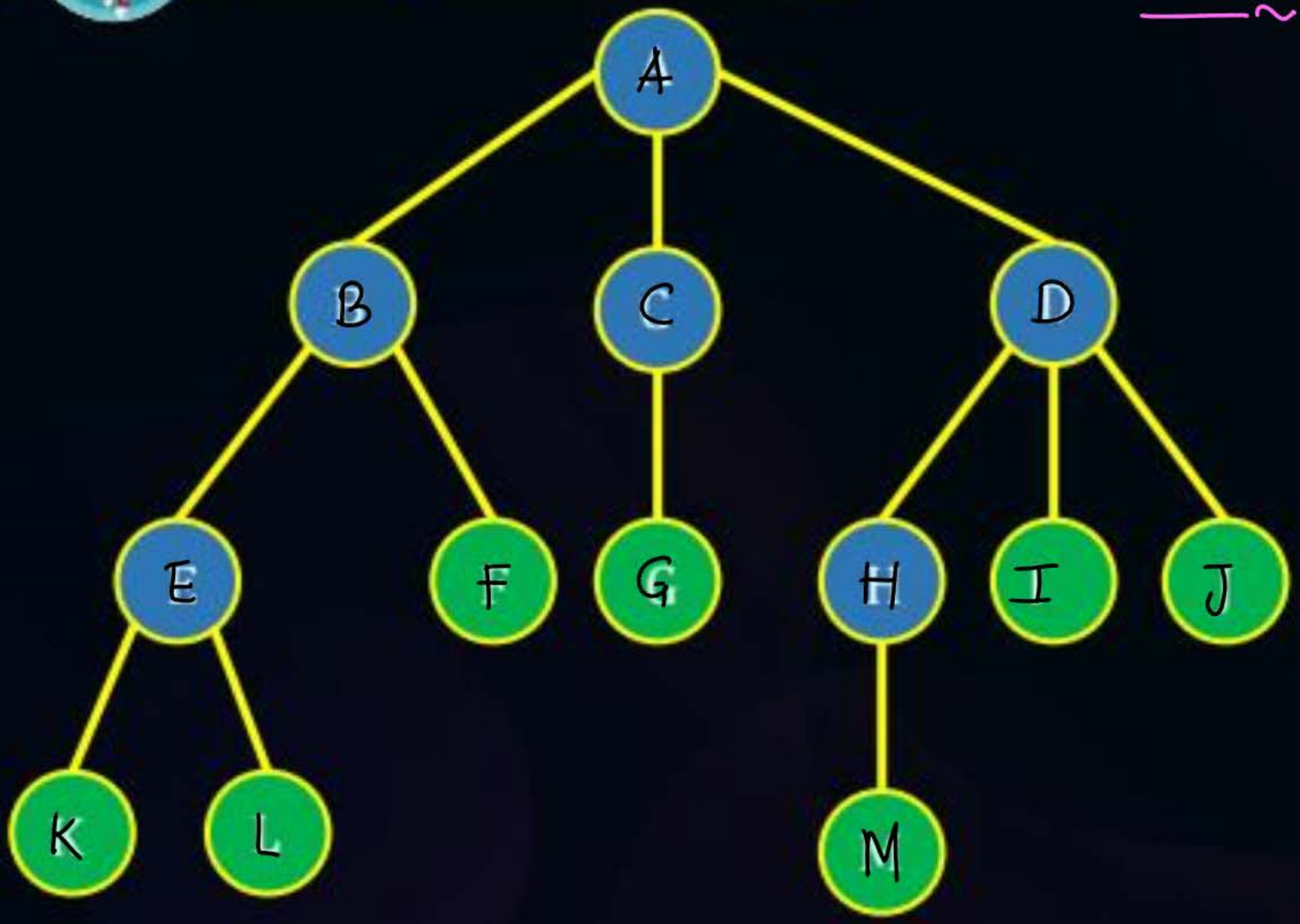
Internal Node :- A node with at least 1 children is internal Node (Non leaf)

Root is having children then its an internal Node

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Topic : Tree



Terminology

L :- No of leaves

I :- No of internal Node

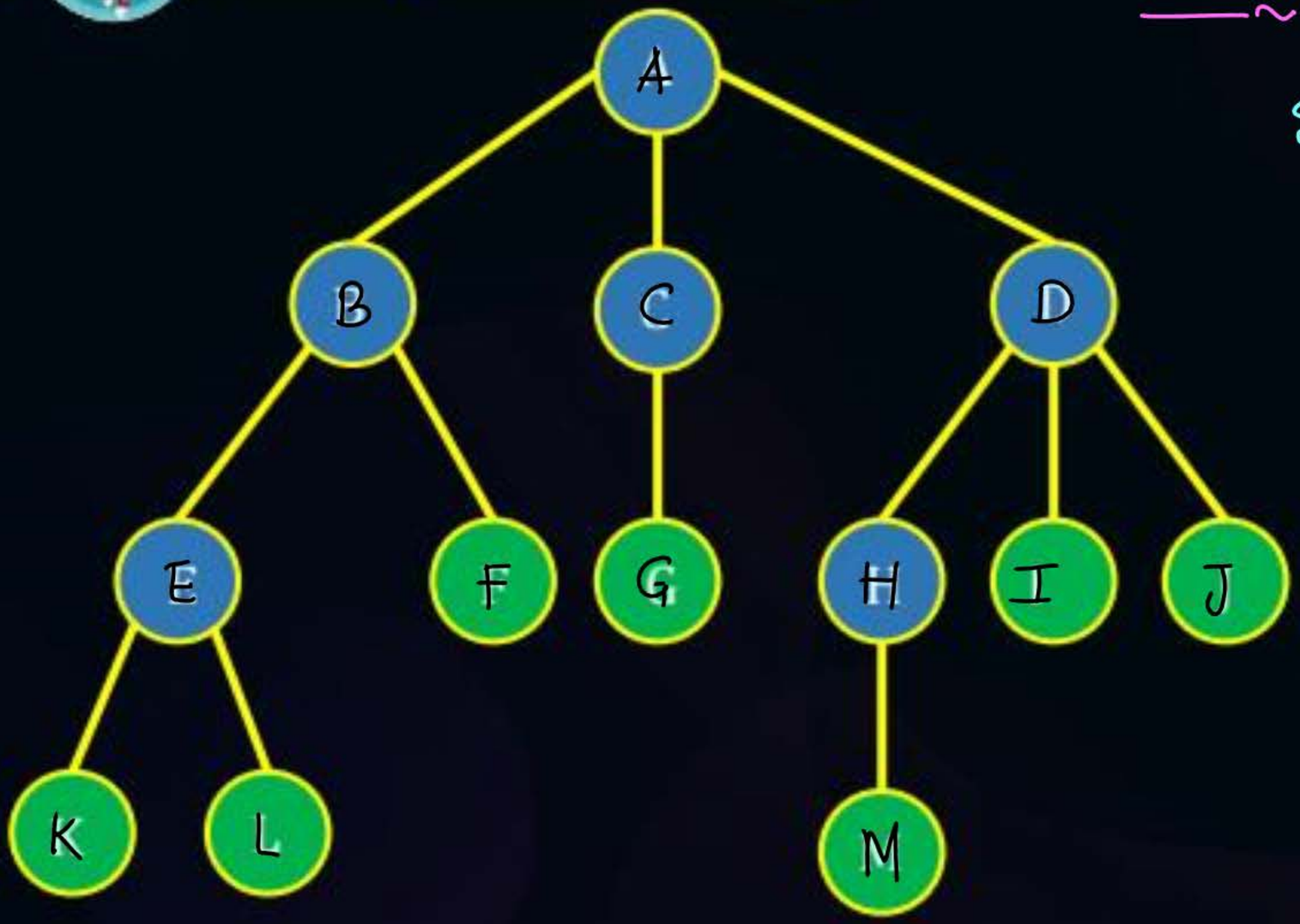
Total No of Node is n

$$n = L + I$$

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Topic : Tree



Terminology

Siblings : children of same parent

called Siblings

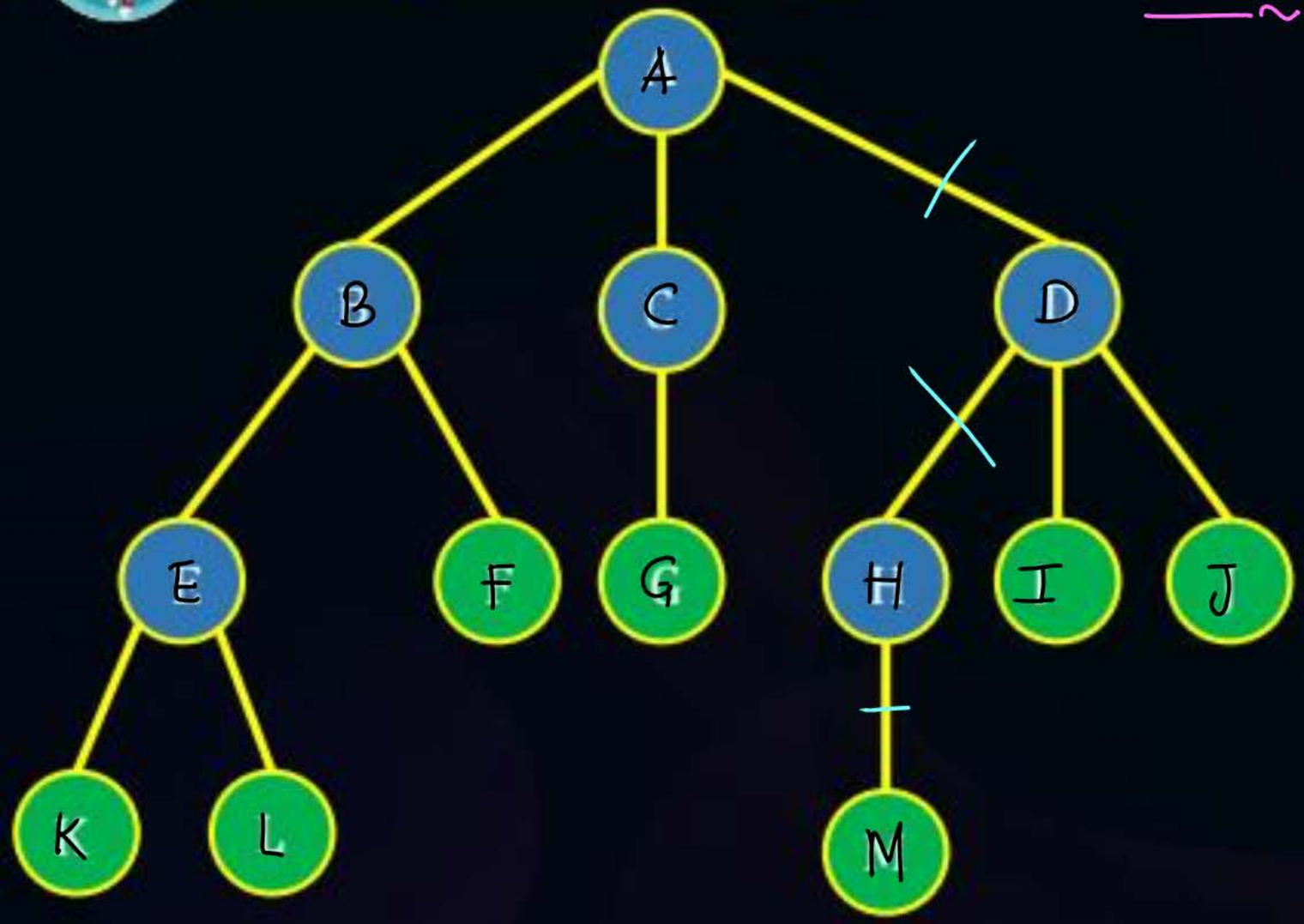
H, I, J, are Siblings

F, G are Not sibling

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Topic : Tree



Terminology

Height of tree is . No. of edges in **Longest** root to Leaf path

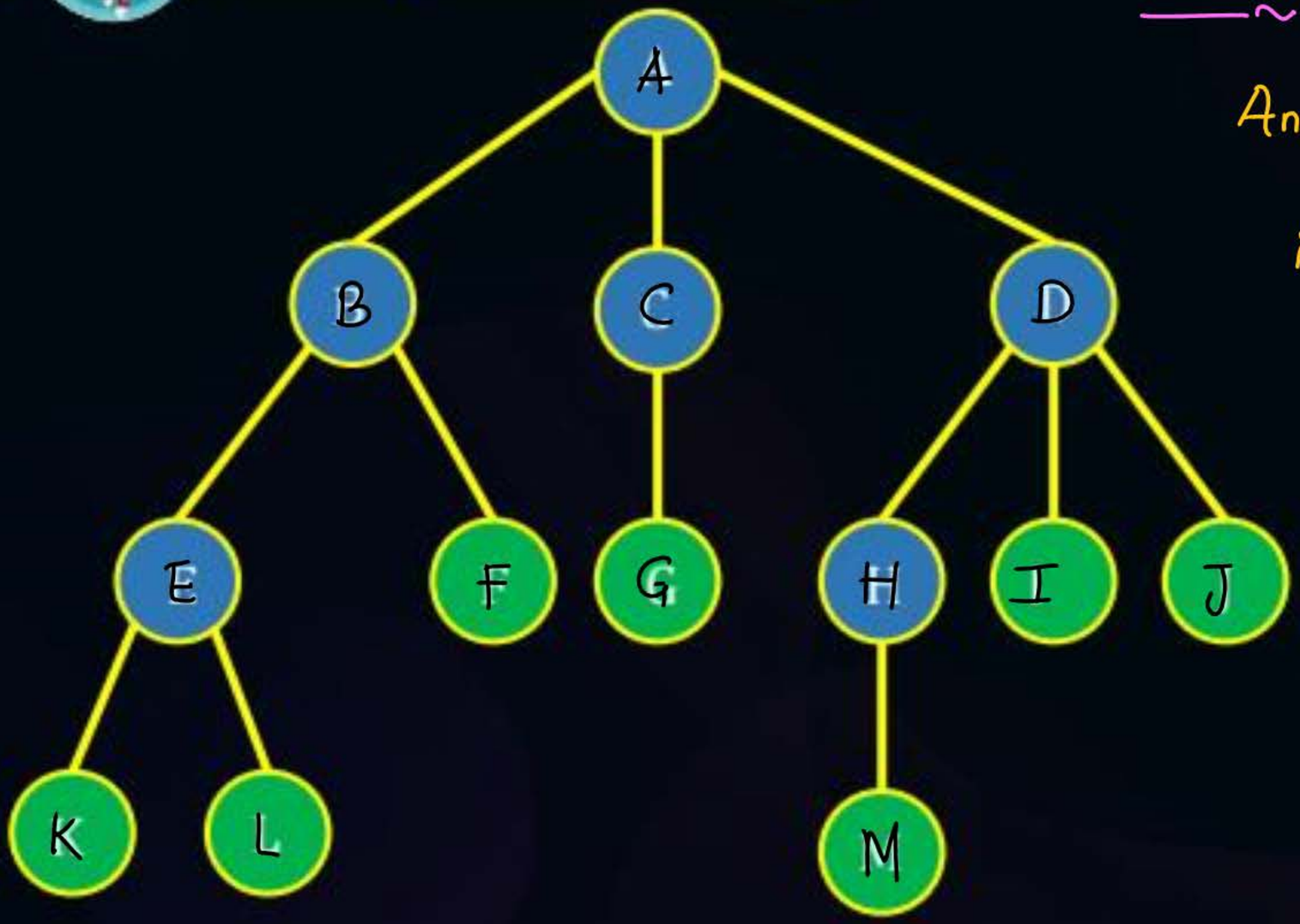
Height of given tree is 3

Depth of a Nucle clefind as No of edges in path of root to the node

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Topic : Tree



Terminology

Ancestor :- Ancestor of a Node x is all node that encounter in Node x to root path.

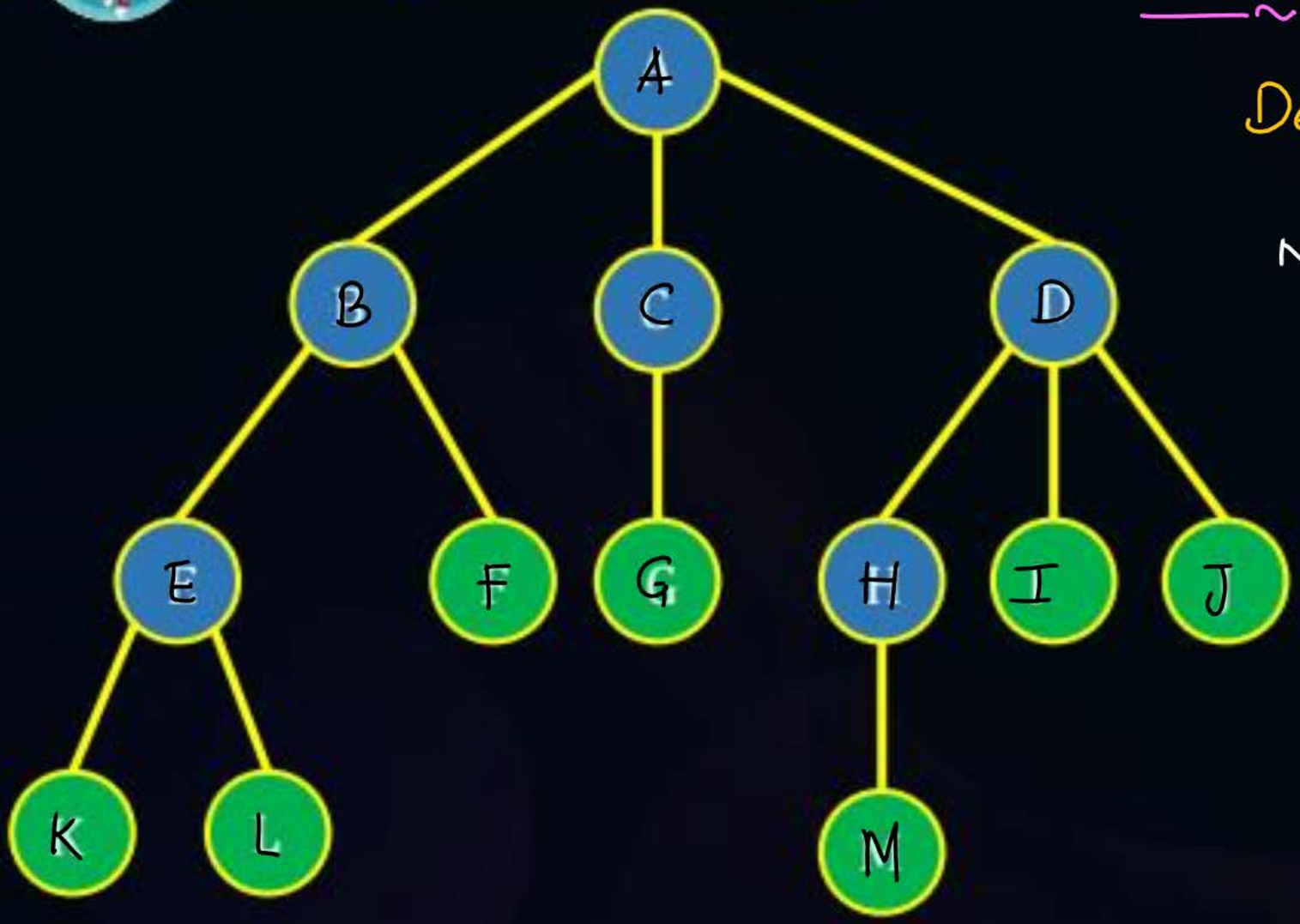
Immediate parent is also considered as ancestor

D, A are ancestor of H

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Topic : Tree



Terminology

Descendant :- Descendant of Node x is all Node that encounter in Node x to leaf path

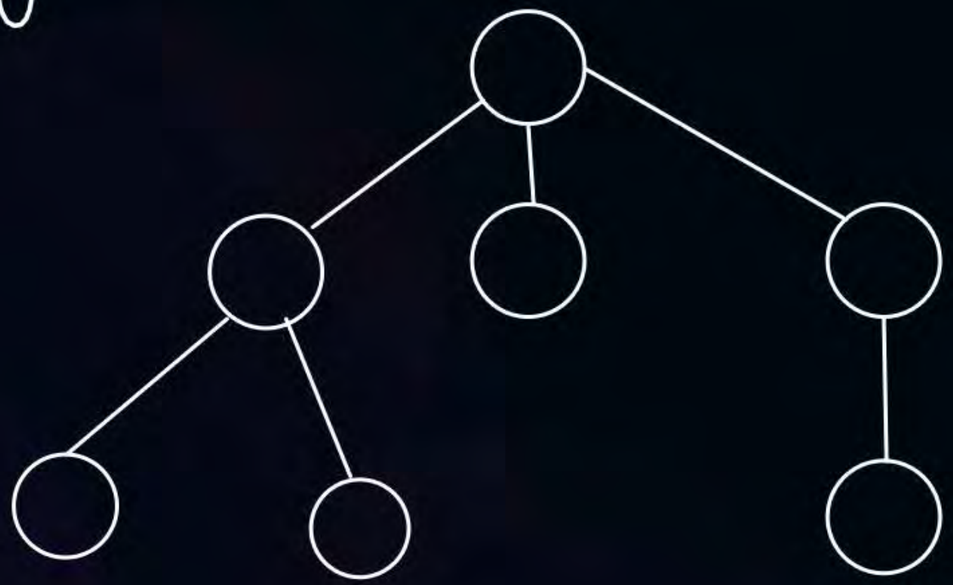
H, I, J, M are descendant of D

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Topic : Tree

General Tree : General tree is a tree in which each node is having 0 or more children



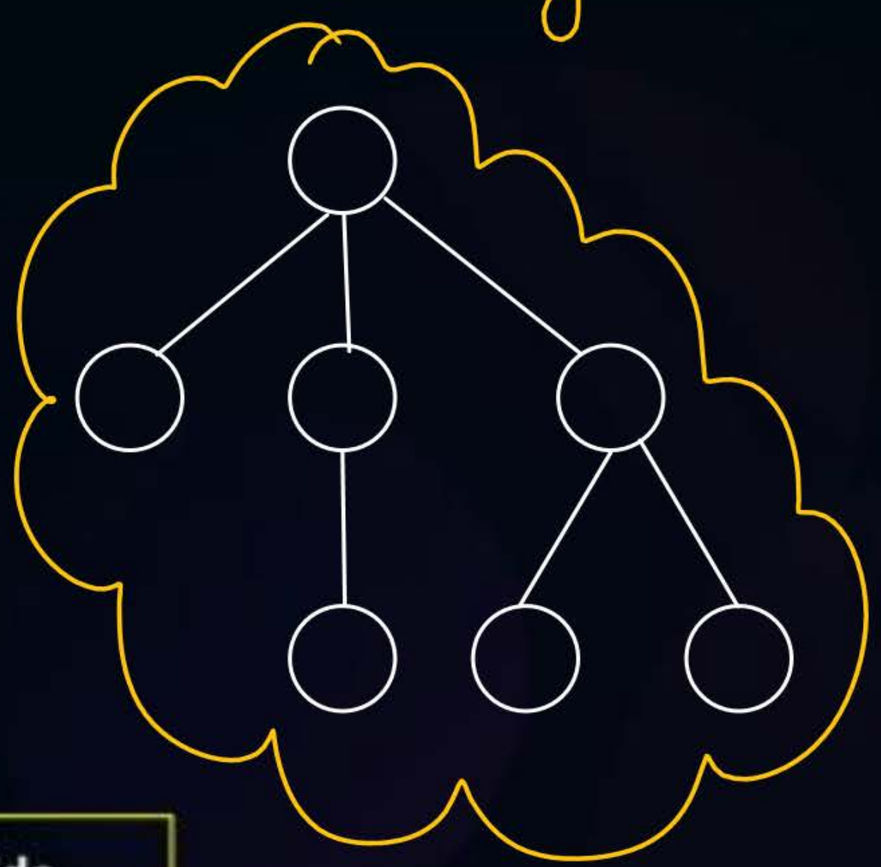
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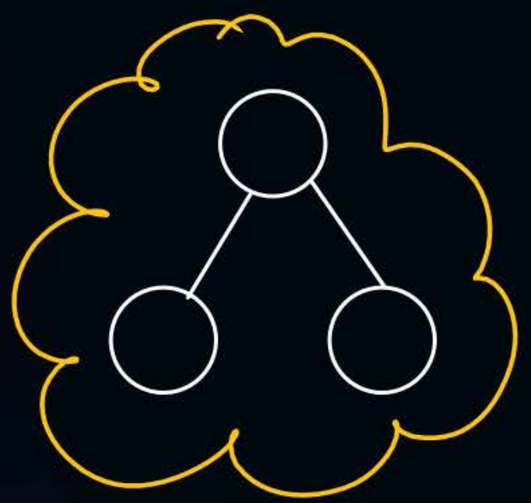


Topic : Tree

forest : forest is collection of one or more general tree.

Each general tree is called component









p = 4

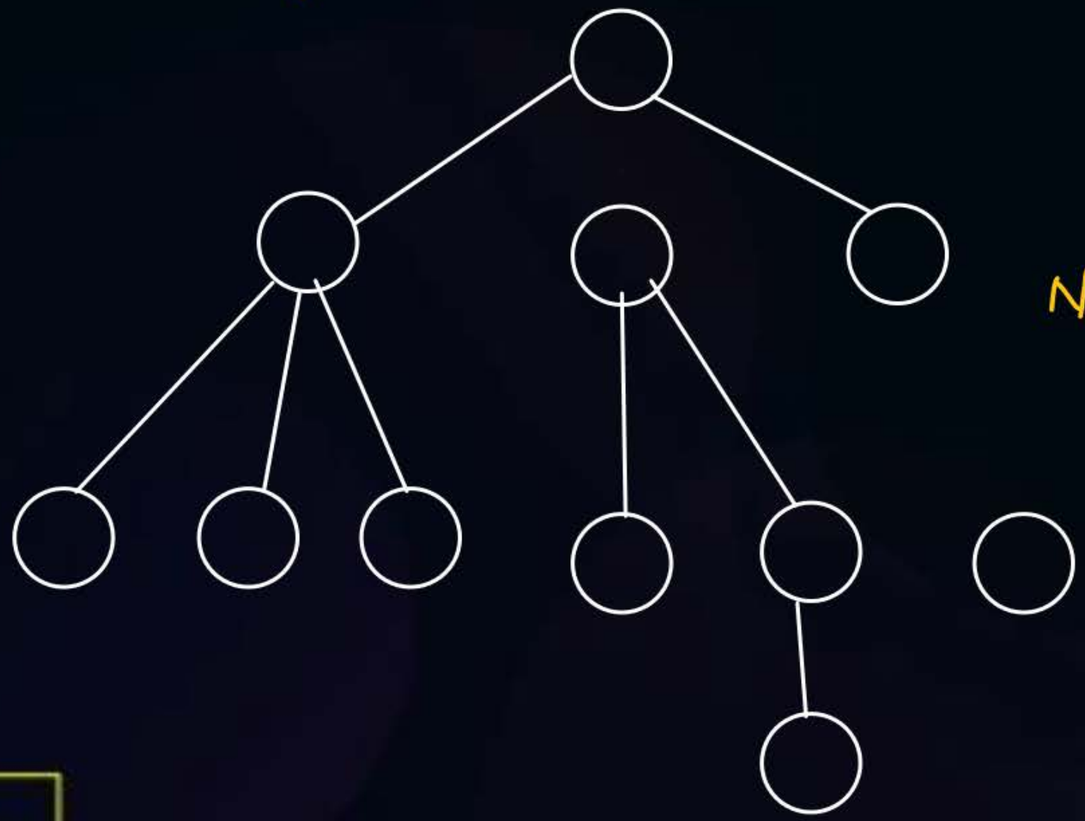
n = 13

No of edges : 9

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Topic : Tree

Theorem : How many edges are present in a forest with n as total No. of Nodes and p as total No. of component. $n-p$



$n = 11$

$p = 1$

No of edges

$11 - 1 = 10$

$n = 11$

$p = 2$

No. of edges

$\therefore$ 9

No. of Nodes : 11

$p = 3$

No. of edges

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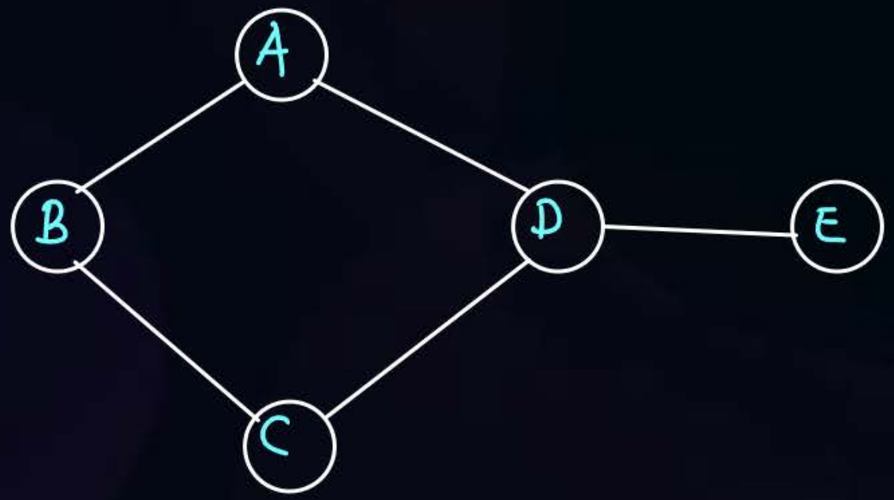
Topic : Tree

Handshaking Algorithm

2. definition

Recent PYQ | By default

degree of a Node defind as No of edges a Node is connected to



degree(A) = 2 , degree(B) = 2

degree(C) = 2 degree(D) = 3

degree(E) = 1

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Topic : Tree

Handshaking Algorithm ∴ Sum of degree is twice the No. of edges

$$\sum_{i=1}^{n} deg(v_i) = 2|E|$$

$2|E| = 2+2+2+3+1 = 10$

$2|E| = 10$

$|E| = 5$



degree(A) = 2 , degree(B) = 2

degree(C) = 2 degree(D) = 3

degree(E) = 1

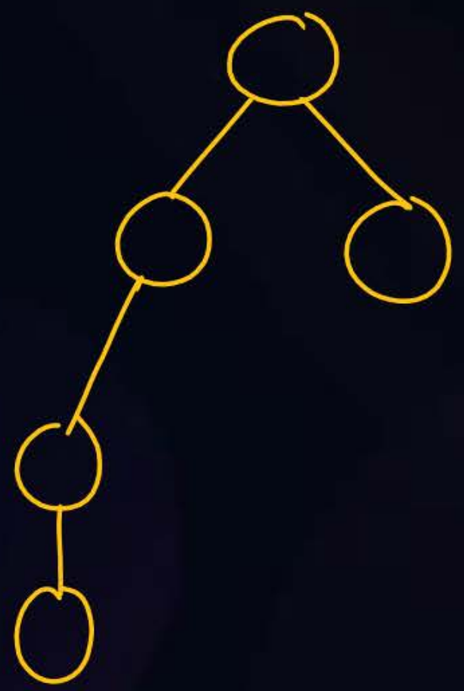
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Topic : Tree

Old PYQ alternative definition of degree in a Tree

degree of a Node in a tree is No. of children a Node is having



No. of Node with degree 0 (0 children)

$n_0 = 2$

No. of Node with degree 1 (1 children)

$n_1 = 2$

No of Node with degree 2 (2 children)

$n_2 = 1$

Slide



Topic : Tree

$n_0 + n_1 + n_2$ = Total No of Node

No. of edges = $n_0 + n_1 + n_2 - 1$

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2 mins Summary

Topic Graph as tree

Topic Rooted tree

Topic General tree, Tree Terminology

Topic forest, forest Theorem

Topic degree of Node

Slide

THANK - YOU

Slide